# Elias Dunham

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## EDUCATION

**Northeastern University, College of Engineering** **Boston, MA**
*Candidate for Bachelor of Science in Mechanical Engineering and Physics* *May 2028*

GPA: 3.8

Coursework**:** Fluid Mechanics, Thermodynamics, Mechanics of Materials, Materials Science, Differential Equations, Physics 2

Honors & Extracurriculars: Dean's List, Commended National Merit Scholar, FBLA State Finalist, American Society of Mechanical Engineers, Northeastern Composing Club, Delta Tau Delta

**Northeastern University, Dialogue of Civilizations** **Panama City, Panama**
*International Applications of Fluid Mechanics* *July 2026*

Studied fluid mechanics via case studies at the Panama Canal, a Penonomé wind farm, and Chiriquí hydroelectric facilities

**Institut Notre-Dame de Bonne-Espérance** **Braine-le-Comte, Belgium**
*Rotary Youth Exchange* *June 2024*

Study abroad program focused on cultural immersion and French language

## COMPUTER & MACHINING SKILLS

**CAD:** SolidWorks, Autodesk Fusion, AutoCAD
**Programming:** C++, JavaScript, Python, MATLAB, Arduino
**Manufacturing:** 3D Printing (SLA, FDM, PμSL), CNC (3- & 5-axis), Manual Injection Molding, Vacuum Compression Molding, Rotary Axis Laser Cutting
**Testing & Instrumentation:** Mechanical Testing (Instron, ADMET), DMM Data Logging, Thermocouple Instrumentation
**Electronics:** PID Control Systems, Oscilloscopes, Soldering, Circuit Engraving (Laser Ablation)

## PROFESSIONAL EXPERIENCE

**MIT Lab for Translational Engineering** **Cambridge, MA**
*Mechanical Engineering Co-op* *January 2026 - Present*

- Lead mechanical design and fabrication across 6 concurrent device programs, taking each from CAD through prototyping and in-vivo validation
- Fabricate device hardware on SLA, PolyJet, and FDM printers, a 5-axis mill, and rotary-axis laser cutter, producing over 50 prototypes to support ongoing device programs
- Establish repeatable fabrication and test protocols, training 4 lab members to operate independently

**FLX Solutions** **Bethlehem, PA**
*Mechanical Engineering Intern* *June - August 2025*

- Programmed a custom application with JavaScript and the OpenAI API to automate company's task assignment protocol
- Trained a Microsoft TrOCR model in Python to convert handwritten notes into searchable database records, digitizing over 1,000 records at 87% accuracy

## PROJECTS

**Wetsuit Thermal Research | US Navy, MIT Lab for Translational Engineering** *March 2026 - Present*

- Engineer and operate a high-pressure testing apparatus with a PID-controlled heated surface (37.2°C) at pressures equivalent to 104 fsw, validating performance in submerged and dry conditions with epoxy-potted heater and thermocouple feedthroughs
- Synthesize and evaluate 24 candidate material formulations, conducting mechanical stress testing with high pressure cycles to characterize structural integrity

**In-Situ Mixing Syringe | MIT Lab for Translational Engineering** *February 2026 - Present*

- Design a dual-barrel syringe in Fusion 360 isolating two components until they mix at the tip, validated through a 23G needle across 8 successful injection trials
- Diagnose leakage from stress concentration at the gasket sealing interface, redistributing sealing load across 11 SLA-printed iterations to eliminate leakage across 8 consecutive trials

**Stem-Cell Delivery Device | MIT Lab for Translational Engineering** *February - June 2026*

- Redesigned the central channel of a stem-cell delivery catheter, reducing taper and flow impedance to prevent cell shearing during *in-vivo* delivery
- Manufactured the complete device sets used for experiments, producing 42 assembled devices and 336 printed and machined parts

**Vagus Nerve Stimulation PCB Housing | MIT Lab for Translational Engineering** *February - April 2026*

- Designed and fabricated a clear SLA housing contoured to the animal's back, converging on assembly tolerances across 15 iterations to keep wire ports and snap-fit lid sealed against fluid ingress throughout 4 in-vivo trials

**Biomimetic Fan Blade Technology | Cornerstone of Engineering** *January - March 2025*

- Developed owl-wing-derived blade geometry in SolidWorks, reaching 30% noise reduction against a baseline design in flow simulation
- Tuned serration amplitude and wavelength against published aeroacoustic data, improving noise reduction to 40% compared to initial prototypes